The Tetrahedron Protocol: Advanced Pedagogical Framework and Technical Fabrication Manual for Distributed Mesh Networks in Primary Education

1. Executive Summary and Pedagogical Architecture

The Tetrahedron Protocol represents a paradigm shift in STEM education, moving beyond screen-based abstraction to the tangible engineering of resilient communication infrastructures. This report outlines a comprehensive curriculum designed to guide junior field agents (students aged 10-14) through the construction, programming, and deployment of a decentralized mesh network node. The project synthesizes three distinct academic disciplines—Euclidean geometry, radio frequency (RF) physics, and computer network topology—into a single, gamified "mission" structure.

The core objective is to demystify the "black box" of modern telecommunications. By constructing a device that transmits data without reliance on centralized cellular towers or internet service providers (ISPs), agents gain a first-principles understanding of connectivity.[1] The curriculum utilizes the "Glass Box" pedagogical approach, where internal mechanisms are exposed and manipulated rather than hidden behind user-friendly abstractions.[2] This builds digital literacy not just in consumption, but in the fundamental architecture of the internet itself.

Central to this project is the use of **Gamification** as a driver for engagement. Research indicates that applying game-design elements—such as "mission briefings," "leveling up," and narrative-driven objectives—significantly increases retention and motivation in complex technical tasks.[3] The "Spy/Agent" narrative provides a cognitive container for the learning, transforming dry safety regulations into "Hazard Protocols" and soldering exercises into "Hardware Fabrication Missions".[3]

This document serves as the definitive manual for the Tetrahedron Protocol. It includes the

"Agent-Facing" documentation (kid-friendly analogies and instructions) embedded within a broader "Instructor-Facing" technical analysis. This dual-layer approach ensures that while the student receives accessible content, the educator is equipped with the deep technical and theoretical background necessary to facilitate the experience.[6]

1.1 The Philosophy of the Tetrahedron

The selection of the tetrahedron as the project's physical and symbolic anchor is deliberate. In geometry, the tetrahedron is the simplest regular convex polyhedron, composed of four triangular faces.[8] It represents the minimum requirements to enclose a three-dimensional space. In network topology, a four-node fully connected mesh graph (K_4) mirrors the vertices of the tetrahedron, where every node has a direct edge to every other node.[9]

This geometric stability translates directly to network resilience. Just as a physical tetrahedron is rigid and resistant to deformation (unlike a cube, which requires internal bracing), a tetrahedral mesh network is resistant to signal interruption. If one link fails, the routing algorithm instantly redirects traffic through the remaining vertices, preserving the integrity of the whole.[1] This synergy between physical form and digital function provides a tactile learning aid for abstract graph theory concepts.

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2. Level 1: Network Topology and Geometric Intelligence

Mission Briefing for Agents:
Welcome, Agent. Before you can build the hardware, you must understand the terrain. The enemy of communication is a broken wire. Your mission is to understand how to build a web that cannot be broken.

2.1 The Spiderweb vs. The Star

To explain complex network topologies to the target demographic, we utilize the **"Spiderweb Analogy"**.[10]

In traditional networking, most students are familiar with the **Star Topology**, even if they do not know the term. This is the model used by home WiFi routers.

- **The Star (Centralized):** Imagine a classroom where the teacher is in the center, and students can only talk to the teacher. If a student wants to pass a note to a friend, they must hand it to the teacher, who hands it to the friend. The teacher is the "Central Node" or "Hub".[1]
    - *Vulnerability:* If the teacher leaves the room (or the router loses power), all communication stops. The network is fragile because it relies on a single point of failure.[1]
- **The Spiderweb (Decentralized Mesh):** Now, imagine the students connect a string from their desk to every other desk. If Student A wants to send a message to Student B, they pull the direct string. If that string snaps, they can hand the note to Student C, who has a string connected to Student B.
    - *Resilience:* This is a **Mesh Network**. There is no central teacher. Every student is a node. Information travels by "hopping" from person to person until it reaches the destination.[10]

Instructor Note on Routing Algorithms:
The kid-friendly "hopping" explanation simplifies the complex operations of routing protocols like OSPF (Open Shortest Path First) or B.A.T.M.A.N. (Better Approach To Mobile Ad-hoc Networking). In a mesh, the device acts as both a receiver and a transmitter (a relay). The "self-healing" capability mentioned in the snippets 1 refers to the algorithm's ability to dynamic recalculate the "cost" of a route. If a node goes offline (infinite cost), the algorithm selects the next lowest-cost path, ensuring data delivery survival.

2.2 The Geometry of the Tetrahedron

The project requires agents to build a chassis in the shape of a tetrahedron. This serves as a physical manipulative to teach Euler's properties of solids.

- **Faces:** 4 (All equilateral triangles).[12]
- **Vertices (Corners):** 4.
- **Edges:** 6.

Mathematical Connection:
The formula for the number of edges in a fully connected graph with n vertices is $E = \frac{n(n-1)}{2}$.
For a tetrahedron ($n=4$):

$$E = \frac{4(3)}{2} = 6$$

This mathematical coincidence means that constructing the physical edges of the tetrahedron chassis perfectly maps to the logical connections in a 4-node mesh network simulation.8

Agent Activity: The String Theory Simulation

To visualize this, agents perform a physical simulation 14:

1. Four agents stand in a circle.
2. They pass a ball of yarn to connect everyone to everyone (creating the 6 edges).
3. The Instructor (representing "Interference") cuts one string.
4. The agents must physically tug on the remaining strings to find a path to the disconnected peer.
5. *Outcome:* This kinesthetic exercise reinforces that while the direct path is gone, the network integrity remains intact through indirect routes.[1]

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3. Level 2: The Physics of Radio (RF) Intelligence

Mission Briefing for Agents:

Agent, your device does not use wires. It uses invisible light called Radio Waves. But not all radio is the same. We do not use the fast, weak voice of the Mouse. We use the slow, deep song of the Whale.

3.1 The "Whale vs. Mouse" Analogy

A critical learning objective is distinguishing between **Bandwidth** (speed) and **Range** (distance). The Tetrahedron Protocol uses **LoRa (Long Range)** technology, which differs fundamentally from the **WiFi** students are used to.

Feature	WiFi (The Mouse)	LoRa (The Whale)
Analogy	A mouse squeaking very fast.	A whale singing a deep, slow song.
Speed	High. Can stream movies and games (100 Mbps+).[15]	Low. Can only send short texts (0.3 - 50 kbps).[16]

Range	Short. Stops at walls. (10-100 meters).[17]	Long. Goes through buildings. (10-15 kilometers).[18]
Energy	High. Drains battery quickly (shouting is tiring).[20]	Low. Lasts for years (whispering is easy).[21]
Frequency	2.4 GHz or 5 GHz (High pitch).[17]	915 MHz or 868 MHz (Low pitch).[21]

Technical Context for Instructors:
The "Whale Song" analogy explains Chirp Spread Spectrum (CSS) modulation.19

- **WiFi (OFDM/DSSS):** WiFi operates at higher frequencies (2.4GHz/5GHz), which behave more like light—they reflect off surfaces and are easily absorbed by water and concrete (high attenuation). To maintain high data rates, they require high Signal-to-Noise Ratios (SNR).[20]
- **LoRa (CSS):** LoRa uses lower frequencies (Sub-GHz) which have longer wavelengths, allowing better penetration of obstacles (diffraction). CSS modulates the signal by varying the frequency over time (a "chirp"). This sweeping signal is mathematically distinct from random background noise, allowing the receiver to decode signals even when they are *below* the noise floor.[22] This is why a LoRa node can "hear" a signal that is indistinguishable from static to a standard radio, much like how a whale's low-frequency call travels across oceans.[22]

3.2 The Spectrum Map

Agents must understand that they are operating in a shared invisible space.

- **The Public Park (ISM Bands):** Just as a park is open to everyone, certain radio frequencies are open for public use without a license. These are the **ISM (Industrial, Scientific, and Medical)** bands.[19]
  - *USA/Americas:* 915 MHz.
  - *Europe:* 868 MHz.
  - *Asia:* 433 MHz.
- **Protocol Violation:** If an Agent sets their radio to a restricted frequency (like the one used by police or airplanes), they are "trespassing." The build guide strictly enforces checking the local regulations to select the correct frequency for the LoRa module.[15]

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4. Level 3: Hazard Protocols and Safety Certification

Security Clearance Required.
Unauthorized personnel are prohibited from entering the Fabrication Zone. All Agents must pass the Safety Briefing to receive their Hardware Kit.
Safety in electronics is not merely about avoiding injury; it is about mindfulness and respect for energy. We reframe safety rules as "Mission Protocols" to ensure compliance through roleplay.[3]

4.1 Protocol: Thermal Defense (Soldering)

The fabrication process involves a soldering iron operating at approx. 400°C (750°F).[23]

The Rules of Engagement:

1. **The "Hot Zone"**: A designated 30cm x 30cm square on the workbench where the iron lives. Nothing flammable (paper, plastic wrappers) enters the Hot Zone. The iron *always* returns to its stand, never the table.[23]
2. **The "Drop Reflex" Override**: Humans have a natural instinct to catch falling objects. Agents must reprogram this instinct. **If the iron falls, let it burn the floor.** Never attempt to catch a falling thermal lance. Jump back and unplug.[24]
3. **Fume Extraction**: The smoke from soldering is not "solder steam"; it is **Flux Fumes** (Rosin). Flux is a chemical acid that cleans the metal. Breathing it can cause "Colophony disease" (occupational asthma) over long periods.[23] Agents must solder in ventilated areas or use fans to push the plume away from their "optical sensors" (eyes) and "intake valves" (nose).[25]
4. **Optical Shielding**: Safety goggles are mandatory. Molten solder can "spit" if it hits moisture. Eyes cannot be replaced.[26]

First Aid Routine (The "Cool Down"):
If a thermal breach (burn) occurs:

- **DO NOT** apply ice (it damages tissue).
- **DO NOT** apply butter or creams (it traps heat).
- **DO** run cool tap water over the breach for 10-15 minutes immediately. Report to Medical Officer (Instructor).[23]

4.2 Protocol: Chemical Containment (LiPo Batteries)

The Tetrahedron Node is powered by a **Lithium Polymer (LiPo)** cell. These high-energy-density cells are safe when respected but volatile if abused.[27]

The "Puff" Test:
Agents are taught to inspect their power cells. A healthy LiPo is flat and firm. A compromised LiPo is swollen or "puffy."

- *Mechanism:* When a LiPo is over-discharged or physically damaged, the electrolyte breaks down and generates gas (mostly hydrogen and oxygen). This inflates the sealed pouch.
- *Directive:* **NEVER** charge a puffy battery. It is a fire hazard. Quarantine it in a sand bucket immediately.[27]

The Short Circuit Hazard:
Touching the red (positive) and black (negative) wires together creates a path of zero resistance.

- *Result:* Massive current dump -> Heat -> Fire.
- *Prevention:* Keep protective caps on connectors until the moment of installation. Never cut both wires at the same time with metal scissors (which bridges the connection).[28]

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5. Level 4: Hardware Fabrication Guide (The Build)

Mission Status: Fabrication Phase.
Objective: Construct the Tetrahedron Node Mk.1. Precision is key. A loose wire is a lost message.

5.1 Bill of Materials (The Armory)

Component ID	Description	Function
MCU-ESP32	Microcontroller (ESP32 or similar)	The "Brain." Runs the code and mesh algorithm.[29]

RF-LORA	LoRa Radio Module (SX1276)	The "Voice." Sends and receives chirps.[19]
ANT-915	915MHz Coil/Wire Antenna	The "Ears." Amplifies weak signals.
HDR-PIN	2.54mm Pin Headers	The "Spine." Connects the Brain to the PCB.[30]
PWR-LIPO	3.7V 500mAh LiPo Battery	The "Heart." Provides energy.[31]
PCB-MAIN	Custom Printed Circuit Board	The "Nervous System." Routes electricity.
CHASSIS	3D Printed Tetrahedron Frame	The "Armor." Protects the core.

5.2 Step 1: The Header Interface (Soldering Basics)

The most critical skill is soldering the pin headers to the microcontroller. This connects the logic to the board.

Technical Theory: Wetting & Intermetallics
Agents often think soldering is like gluing. It is not. Glue sits on top. Solder fuses into the metal.

- *The process:* When the iron heats the copper pad and the leg of the component, the **Flux** melts first, cleaning off invisible oxidation.[23] Then the solder melts and flows. It forms an **Intermetallic Compound**—a new alloy where the copper and tin mix at the atomic level.
- *The "Bad" Joint:* If the pad isn't hot enough, the solder balls up on top (Cold Joint). It makes no electrical connection.

Operational Guide:

1. **Stabilize**: Insert the long legs of the header pins into a breadboard to hold them straight. Place the Microcontroller on top.[30]
2. **Heat**: Touch the iron tip to **BOTH** the metal pin and the gold ring (pad) on the board.
3. **Count**: "One Mississippi, Two Mississippi..." (This heats the metal).
4. **Feed**: Touch the *solder wire* to the joint (not the iron). It should melt and flow like shiny

silver water around the pin.[32]

5. **Withdraw**: Remove the wire. Then remove the iron.
6. **Freeze**: Wait 3 seconds. Do not blow on it. The joint should look like a shiny **volcano** or **Hershey's Kiss**.[33] If it looks like a ball, re-heat it.

5.3 Step 2: The Radio Integration

The LoRa module connects to the microcontroller via the SPI (Serial Peripheral Interface) bus.

- **Pin Mapping**:
  - **MOSI** (Master Out Slave In) -> Data from Brain to Radio.
  - **MISO** (Master In Slave Out) -> Data from Radio to Brain.
  - **SCK** (Serial Clock) -> Keeps them in sync.
  - **CS/NSS** (Chip Select) -> Wakes up the radio.[34]
- **Action**: Solder the radio module to the PCB. Ensure the orientation matches the silk-screen markings. A reversed radio is a dead radio.[35]

5.4 Step 3: Antenna Tuning

The antenna length is determined by physics, specifically the wavelength of the frequency.

$$\lambda = \frac{c}{f}$$

- c = Speed of Light (300,000 km/s).
- f = Frequency (e.g., 915 MHz).
- For 915 MHz, a full wave is ~33cm. We typically use a **Quarter Wave Monopole**, which is $\frac{1}{4}$ of that length (~8.2 cm).
- **Directive**: Do not cut the antenna wire arbitrarily. Precision to the millimeter ensures the "Whale Song" is in tune.[19]

5.5 Step 4: Chassis Assembly

1. **The Base**: Mount the PCB to the bottom triangular face using nylon standoffs. Metal screws can cause short circuits.[36]

2. **The Geometry**: Fold the three side faces up to meet at a single point (vertex).
3. **The Enclosure**: Secure the edges. Ensure the antenna is not kinked or touching metal parts, which would detune it (standing wave ratio mismatch).

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6. Level 5: Deployment and Network Topography

Mission Status: Active.
Objective: Establish the Mesh. Visualize the invisible.

6.1 The "Gossip" Protocol (Flooding)

How does a message get from Node A to Node D if they can't see each other? The software uses a controlled **Flooding Algorithm**, often called "Gossip."

1. **Origin**: Node A shouts a message ("Packet").
2. **Relay**: Node B hears it. It checks its "Diary" (Memory).
   - *Has it seen this Packet ID before?*
   - **Yes**: Ignore it (prevent echo).
   - **No**: Add to Diary and Shout it out again.
3. **Propagation**: The message ripples outward like a wave in a pond until it hits every node, including destination D.[1]

6.2 Field Exercise: The Range Finder (RSSI Mapping)

To test the "Whale Song," agents engage in an **RSSI (Received Signal Strength Indicator)** Hunt.

- **The Metric**: RSSI is measured in dBm (decibel-milliwatts). It is a negative number.
  - **-30 dBm**: Extremely strong (Same room).
  - **-90 dBm**: Weak (Down the street).
  - **-120 dBm**: The limit of reception (The "Noise Floor").[22]

Game: The Fox Hunt

1. **The Fox**: One node is hidden in the schoolyard, transmitting a beacon every 5 seconds.
2. **The Hunters**: Agents use their handheld nodes to find the Fox.

3. **The Strategy**: Agents must watch the RSSI number. If it moves from -95 to -80, they are getting "Warmer." If it drops to -100, they are getting "Colder." This teaches triangulation and radio propagation characteristics.[14]

6.3 Field Exercise: The Faraday Cage Challenge

Agents are challenged to "block" the unblockable signal.

1. Place a node inside a cardboard box. Signal passes (RSSI changes slightly).
2. Place a node inside a metal cookie tin or wrap it in aluminum foil.
3. *Observation:* Signal dies immediately.
4. *Theory:* This is a **Faraday Cage**. The metal conducts the radio waves around the outside of the box, cancelling the field inside. This demonstrates why elevators kill cell phone signals.[37]

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7. Level 6: Troubleshooting and Maintenance (The Repair Bay)

System Alert: Anomaly Detected.
Engineering is not about building things that work; it is about fixing things that break.
Debugging is the highest form of logic.

7.1 The Binary Search Debugging Method

We teach agents to solve problems using the **"Split the World"** technique (Binary Search).[38]
Do not guess. Divide the problem space.

- **Problem:** The screen is dark.
- **Split 1:** Is it Hardware or Software?
  - *Test:* Plug it into power. Does the "Charge" LED light up?
  - *Result Yes:* Hardware is receiving power. The problem is likely the screen or code.
  - *Result No:* Hardware is dead. Check battery and switch.

7.2 The "Smoke Test"

The most primitive but effective diagnostic.

- **Procedure:** Power on the device. Sniff.
- **Symptom:** Smell of burning plastic or ozone ("Magic Smoke").
- **Diagnosis:** Short Circuit. Immediate Shutdown.
- **Fix:** Inspect solder joints for "bridges" (solder connecting two pins that shouldn't touch). Use a solder sucker to remove excess.[35]

7.3 Debugging Flowchart (Narrative Version)

1. **Power Check**: Is the "Heartbeat" LED blinking?
   - *No:* Check Battery JST connector. Check battery voltage (>3.5V).
2. **Radio Check**: Is the "LoRa Init" message on screen?
   - *No:* "Radio Failure." Check the soldering on the SPI pins (MOSI, MISO, SCK). The brain cannot talk to the voice.[34]
3. **Comms Check**: Can you see other nodes?
   - *No:* Check Frequency settings in code. Are you on 915MHz while they are on 433MHz? Check Antenna connection.[19]

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8. Conclusion and Future Outlook

The Tetrahedron Protocol is more than a soldering workshop; it is an initiation into the infrastructure of the future. By completing this curriculum, Agents have traversed the history of communication—from the structural stability of the tetrahedron to the spread-spectrum resilience of modern IoT networks.

They have learned that:

1. **Resilience is Topological**: A distributed mesh withstands failure better than a centralized star.[10]
2. **Physics is Negotiable**: We can trade speed for range (LoRa vs. WiFi) to solve different problems.[20]
3. **The "Black Box" is Open**: The internet is not magic; it is a series of radios, wires, and protocols that they now have the power to build, repair, and improve.[2]

As we move toward a world of the **Internet of Things (IoT)**, where billions of devices will communicate autonomously, the skills forged here—circuit fabrication, network logic, and spectrum awareness—will be the literacy of the 21st century.[17] The mesh is now active. The mission continues.

End of Report.

Appendix A: Detailed Safety Protocol Citations and References

A.1 Soldering Safety Standards

- **Source**: *University Health & Safety Guidelines for Soldering*.[23]
- **Key Data**: Melting point of solder (approx 188°C-217°C for lead-free), Iron tip temp (350°C-400°C).
- **Risk Mitigation**:
  - **Inhalation**: Rosin flux fumes are a sensitizer. Use HEPA fume extractors or cross-ventilation.[23]
  - **Contact**: Third-degree burns occur instantly at 400°C. Cold water treatment is mandated over ice to prevent vasoconstriction and tissue damage.[23]
  - **Ingestion**: Lead residues (if using leaded solder) require hand washing before eating. Lead-free solder is recommended for K-12 environments.[39]

A.2 Lithium Polymer (LiPo) Handling

- **Source**: *Remote Control Hobbyist Safety Standards*.[27]
- **Key Data**: Nominal voltage 3.7V. Max charge voltage 4.2V. Min discharge voltage 3.0V.
- **Risk Mitigation**:
  - **Thermal Runaway**: Caused by physical puncture or overcharging. Extinguish with sand (Class D fire agent) or chemical extinguisher. Water is generally ineffective on the chemical reaction but cools the surrounding area.[40]
  - **Storage**: Store at "Storage Voltage" (3.8V per cell) in fireproof bags.[27]

A.3 RF Exposure and Compliance

- **Source**: *FCC Part 15 / ETSI EN 300 220*.[15]
- **Key Data**: ISM Bands (915 MHz US / 868 MHz EU).
- **Risk Mitigation**:
  - **Duty Cycle**: LoRaWAN regulations often impose a 1% duty cycle (can only transmit 1% of the time) to prevent jamming the public airwaves. The Tetrahedron Protocol software limits transmission rate to comply with these "fair use" rules.[18]

Appendix B: Gamification Strategy Analysis

B.1 The Role of Narrative in STEM

Using the "Spy/Agent" persona acts as a scaffolding mechanism.

- **Cognitive Load**: Complex technical terms (Circuit Board, Microcontroller) are mapped to familiar narrative concepts (The Map, The Brain), reducing anxiety and cognitive load.[3]
- **Motivation**: "Mission Briefings" create extrinsic motivation that transitions into intrinsic motivation as the agent succeeds in building functional tech.[7]
- **Failure Reframing**: In a game, failure is just "Game Over, Try Again." In a classroom, failure is a bad grade. By framing troubleshooting as "Decoding the Anomaly," we encourage persistence (Grit).[6]

B.2 Mission Progression Table

Mission Level	Educational Goal	Gamified Title	Artifact Produced
Lvl 1	Topology & Graph	"The Spider's Web"	String Network Map

	Theory		
Lvl 2	Soldering & Assembly	"Forging the Artifact"	Functional PCB
Lvl 3	Code & Firmware	"The Awakening"	Flashing LED (Heartbeat)
Lvl 4	Field Testing	"The Fox Hunt"	RSSI Range Data
Lvl 5	Encrypted Messaging	"The Enigma"	Encrypted Mesh Chat

This structured progression ensures that agents experience "Small Wins" at every stage, maintaining the dopamine loop necessary for sustained engagement in difficult engineering tasks.[3]

Works cited

1. Mesh networking - Wikipedia, accessed November 28, 2025, https://en.wikipedia.org/wiki/Mesh_networking
2. DIY retro games console teaches kids about electronics - BBC Click - YouTube, accessed November 28, 2025, https://www.youtube.com/watch?v=N4Ye-5W0PVQ
3. 8 Ways to Gamify Your Classroom Instruction - Discovery Education, accessed November 28, 2025, https://www.discoveryeducation.com/blog/teaching-and-learning/8-ways-to-gamify-your-classroom-instruction/
4. 10 Gamification in Education Ideas to Make Learning Fun - University of San Diego Professional & Continuing Ed, accessed November 28, 2025, https://pce.sandiego.edu/gamification-in-education/
5. TOP SECRET MISSIONS AWAIT! - YMI Classroom, accessed November 28, 2025, https://ymiclassroom.com/wp-content/uploads/2022/06/spy_activitieskit.pdf
6. Creating the Game of Games (a Sample Digital Learning Resources Project) - YouTube, accessed November 28, 2025, https://www.youtube.com/watch?v=rgNWtinUp_s
7. Gamification for Learning: 7 Strategies to Enhance Student Motivation & Achievement, accessed November 28, 2025, https://centrical.com/resources/blog-gamification-for-learning/
8. Tetrahedron - Wikipedia, accessed November 28, 2025, https://en.wikipedia.org/wiki/Tetrahedron
9. accessed November 28, 2025,

https://study.com/academy/lesson/tetrahedron-definition-formula.html#:~:text=Regular%20Tetrahedron,-A%20regular%20tetrahedron&text=The%20interior%20angles%20of%20an,edges%20are%20of%20equal%20length.
10. Mesh networking Facts for Kids, accessed November 28, 2025, https://kids.kiddle.co/Mesh_networking
11. Understanding Mesh WiFi: How It Works and Why You Need It - YouTube, accessed November 28, 2025, https://www.youtube.com/watch?v=ofDUhiY8q0w
12. Meaning, Properties, Formulas | Tetrahedron Shape - Cuemath, accessed November 28, 2025, https://www.cuemath.com/geometry/tetrahedron/
13. Tetrahedron | Definition, Examples, Properties, Nets & Formulas - Helping with Math, accessed November 28, 2025, https://helpingwithmath.com/tetrahedron/
14. Blast Off Learning: Your Ultimate Mars Rover STEM Activity Guide! | I'm the Chef Too, accessed November 28, 2025, https://www.imthecheftoo.com/blogs/stem-for-kids/blast-off-learning-your-ultimate-mars-rover-stem-activity-guide
15. What's The Difference Between LoRa And WIFI？Spotlight On Wireless Communication Connectivity - IOTROUTER, accessed November 28, 2025, https://en.iotrouter.com/spotlight-on-wireless-communication-connectivity-whats-the-difference-between-lora-and-wifi/
16. How LoRa works and what makes it better than Wi-Fi and LTE - Device Prototype, accessed November 28, 2025, https://deviceprototype.com/en/how-lora-works-and-what-makes-it-better-than-wi-fi-and-lte/
17. WiFi and LoRaWAN®: Friends or Foes? - Comcast Business, accessed November 28, 2025, https://business.comcast.com/community/browse-all/details/wifi-and-lorawan-friends-or-foes
18. Finally understandable: LoRa (and LoRaWAN) explained simply! - LineMetrics, accessed November 28, 2025, https://www.linemetrics.com/en/finally-understandable-lora-and-lorawan-explained-simply/
19. Introducing LoRa™ ! : 19 Steps - Instructables, accessed November 28, 2025, https://www.instructables.com/Introducing-LoRa-/
20. LoRaWAN vs. Wi-Fi - FSG Smart Buildings, accessed November 28, 2025, https://www.fsgsmartbuildings.com/lorawan-vs-wi-fi/
21. LoRa Module vs. WiFi: Why LoRa is the Best Choice for Long-Range, Low-Power IoT, accessed November 28, 2025, https://www.nicerf.com/news/lora-module-vs-wifi-which-is-best.html
22. What is LoRa: The fundamentals. It is not a protocol. Not an audio... | by Vit Prajzler | Medium, accessed November 28, 2025, https://medium.com/@prajzler/what-is-lora-the-fundamentals-79a5bb3e6dec
23. Safe Soldering Work Practices, accessed November 28, 2025, https://www.hvcc.edu/ehs/health/safe-soldering-work-practices.pdf
24. You should teach your kids to solder | by Brandon Satrom | Breadboardeaux - Medium, accessed November 28, 2025,

https://medium.com/breadboardeaux/you-should-teach-your-kids-to-solder-45de6c15385b
25. STEM Activity for Teens: Simple Safety Tips to Solder for Beginners, accessed November 28, 2025, https://www.survivingateacherssalary.com/stem-activity-teens-simple-safety-tips-solder-beginners/
26. Soldering iron Facts for Kids, accessed November 28, 2025, https://kids.kiddle.co/Soldering_iron
27. Lipo Battery Safety is IMPORTANT! - Hobby Works, accessed November 28, 2025, https://www.hobbyworks.com/ccms/index.cfm/articles/lipo-battery-safety-is-important/
28. Lipo and Li-Ion Battery Safety Guide - Mr Positive, accessed November 28, 2025, https://www.mrpositive.co.nz/buying/knowledge-base/lipo-and-li-ion-battery-safety-guide/
29. How to Solder Pins Header : 7 Steps (with Pictures) - Instructables, accessed November 28, 2025, https://www.instructables.com/How-to-Solder-Pins-Header/
30. How To Solder Headers - Adafruit Learning System, accessed November 28, 2025, https://learn.adafruit.com/how-to-solder-headers?view=all
31. Safely Using LiPo Batteries - Charging, Balancing & Storing - DroneBot Workshop, accessed November 28, 2025, https://dronebotworkshop.com/lipo-safety/
32. How to solder pin headers - YouTube, accessed November 28, 2025, https://www.youtube.com/watch?v=TFfMYYBRu30
33. How To Solder Headers: A Beginner Tutorial #adafruit - YouTube, accessed November 28, 2025, https://www.youtube.com/watch?v=Z0joOKaQ43A
34. Troubleshooting a Microcontroller Project - Dr. P's Engineering Classes, accessed November 28, 2025, https://doctor-pasquale.com/2020/09/21/troubleshooting-a-microcontroller-project/
35. Troubleshooting electronic projects - BuildCircuit.COM, accessed November 28, 2025, https://www.buildcircuit.com/troubleshooting-electronic-projects/
36. Creating your own Handheld Electronic Games – Pocket Nim - element14 Community, accessed November 28, 2025, https://community.element14.com/challenges-projects/project14/electronic-toys/b/blog/posts/creating-your-own-handheld-electronic-games-pocket-nim
37. Why use LoRa instead of WiFi? - LoRa Antenna, accessed November 28, 2025, https://www.loraantenna.com/why-use-lora-instead-of-wifi/
38. Flowchart for troubleshooting a computer problem - Visual Paradigm Online, accessed November 28, 2025, https://online.visual-paradigm.com/diagrams/templates/flowchart/flowchart-for-troubleshooting-a-computer-problem/;VPSESSIONID=3162BF9685BB0DF4079F749021667EB1
39. Soldering Safety - Department of Engineering Health & Safety | - University of Cambridge, accessed November 28, 2025, https://safety.eng.cam.ac.uk/safe-working/copy_of_soldering-safety
40. Safely Using LiPo Batteries - YouTube, accessed November 28, 2025,

https://www.youtube.com/watch?v=7Gh1f-hBLOU
41. Educational Toys for 8 Year Olds Your Child Will Love - Space Ranger Fred, accessed November 28, 2025, https://www.spacerangerfred.com/2025/11/24/educational-toys-for-8-year-olds/
42. A Preliminary Analysis of Gamification on Assembly Language Programming Education - Academic Conferences International, accessed November 28, 2025, https://papers.academic-conferences.org/index.php/ecgbl/article/download/3886/3738/14927
43. Gamify Any Learning with ThingLink's New Locked Mode, accessed November 28, 2025, https://www.thinglink.com/blog/gamify-any-learning-with-thinglinks-new-locked-mode/